KRISTINE STANLEY

PYTHON / FLASK BACKEND ENGINEER

Coaldale, CO 81222

+1 (786) 949-7561 | kristine.stanley.dev@outlook.com | linkedin.com/in/kristine-stanley-341688411

PROFESSIONAL SUMMARY

Senior Python / Flask Backend Engineer with 11+ years across cybersecurity, connected mobility, insurance, and learning platforms, delivering Python 3.8-3.12, Flask 2-3, PostgreSQL 12-16, Kafka 2-3 solutions with secure APIs, scalable architecture, cloud delivery, observability, and production modernization that improves reliability, performance, release quality, and business scalability for enterprise products.

SKILLS

- **Backend Engineering:** Python 3.8-3.12, Flask 2-3, SQLAlchemy, PostgreSQL, Redis, Celery, Kafka, AWS, Docker, distributed services, API design, background jobs, transactional workflows, performance tuning, secure integrations
- **API & Security:** REST, GraphQL, gRPC, OpenAPI, OAuth 2.0, OpenID Connect, JWT, RBAC, rate limiting, idempotency, webhooks, service contracts
- **Architecture:** clean architecture, domain-driven design, microservices, event-driven architecture, modular monoliths, CQRS, transactional outbox, API versioning
- **Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, Terraform, Helm, GitHub Actions, Azure DevOps, Jenkins, CI/CD, feature flags, blue-green deployments
- **Databases & Messaging:** PostgreSQL, SQL Server, Redis, MySQL, MongoDB, Kafka, RabbitMQ, Azure Service Bus, AWS SQS, indexing, migrations, query optimization, caching
- **Testing & Quality:** xUnit, NUnit, Playwright, Cypress, Jest, Testcontainers, Pact, integration testing, contract testing, load testing, accessibility testing, static analysis, secure code review
- **Observability & Leadership:** OpenTelemetry, Serilog, Datadog, Application Insights, Prometheus, Grafana, incident response, root-cause analysis, mentoring, architecture reviews, stakeholder communication

RELEVANT EXPERIENCE

Senior Software Engineer
April 2021 - May 2026

Veracode Systems Inc — Remote / Full time, Somerville, MA

- Led senior delivery for enterprise application-security products using **Python 3.8-3.12, Flask 2-3, SQLAlchemy, PostgreSQL, Redis, Celery, Kafka, Docker, Kubernetes** across secure APIs, distributed services, background workers, reporting workflows, and operational integrations for security, product, platform, and support teams.
- Reorganized tightly coupled services into domain-focused modules using **clean architecture**, typed contracts, dependency injection, and reviewable service boundaries, reducing recurring production defects by **36%**.
- Resolved P95 latency caused by repeated aggregate queries, oversized payloads, blocking network calls, and inefficient database access patterns through **query tuning**, caching, pagination, asynchronous processing, and payload shaping, improving critical workflow response time by **43%**.
- Modernized legacy components from older runtime patterns to **Python Backend** practices, replacing obsolete packages, synchronous calls, unsafe shared state, and fragile deployment assumptions with maintainable release paths.
- Designed secure multi-tenant access patterns with **OAuth 2.0**, **OpenID Connect**, JWT validation, tenant-scoped data access, RBAC, audit logging, and policy-based authorization for sensitive customer data.
- Eliminated duplicate processing during retries by implementing **idempotency keys**, transactional state changes, outbox publishing, dead-letter replay procedures, and correlation identifiers across service boundaries.
- Built workflow APIs, admin tools, integration endpoints, event consumers, and secure customer-facing capabilities while coordinating backlog priorities, technical design, rollout planning, and dependency management with product managers, QA engineers, security reviewers, and customer-facing teams.
- Established automated quality gates with **unit tests, integration tests, contract tests, migration checks, load tests, and service virtualization**, catching contract mismatches, migration failures, data regressions, and release-blocking defects before production deployment.
- Introduced operational dashboards with **OpenTelemetry, Serilog, Datadog, Application Insights, Prometheus, Grafana**, accelerating diagnosis of memory leaks, queue backlogs, database contention, failed integrations, and downstream timeout incidents.

Software Engineer
January 2019 - April 2021

Cubic Telecom USA — Remote / Full time, San Diego, CA

- Developed connected-mobility platform services using **Python 3.7-3.9, Flask 2-3, PostgreSQL, RabbitMQ, Celery, Azure DevOps** for device registration, SIM provisioning, subscription changes, carrier synchronization, billing events, and customer operations.
- Improved provisioning throughput by **38%** through asynchronous command handling, bounded concurrency, batched persistence, rate limiting, and resilient retry policies that protected external telecom APIs.
- Resolved failed activations caused by inconsistent carrier identifiers, incompatible payload structures, and delayed callbacks by creating canonical models, provider adapters, and replayable reconciliation workflows.
- Designed versioned API contracts with **OpenAPI**, validation rules, standardized error responses, permission-aware endpoints, and documented integration examples for portals and operational tools.
- Implemented message-driven workflows with durable queues, dead-letter handling, consumer checkpoints, idempotent updates, and correlation identifiers to prevent duplicate activations and preserve traceability.
- Corrected stale device states caused by out-of-order events through timestamp validation, monotonic transition rules, scheduled reconciliation jobs, and immutable status histories for support teams.
- Optimized slow device, account, and usage queries using covering indexes, compiled projections, connection-pool tuning, and removal of N+1 access patterns as connected-device volumes increased.
- Standardized delivery with **Docker**, **Azure DevOps**, automated tests, security scans, migration validation, environment checks, approval gates, and rollback notes for distributed releases.

Software Developer
January 2017 - December 2018

Asurion — Remote / Full time, Nashville, TN

- Built insurance and service-contract workflows using **Python 3.5-3.7, Flask, C# integrations, SQL Server, Redis, REST services** for policy eligibility, claim intake, payment callbacks, appointment scheduling, repair status, and customer communications.
- Increased successful online claim completion by **27%** after redesigning orchestration flows to preserve recoverable state, return actionable validation details, and reduce unnecessary calls to slow dependencies.
- Created a backend-for-frontend and integration layer that combined customer, policy, inventory, payment, and appointment data while shielding product experiences from incompatible downstream contracts.
- Corrected duplicate appointment and payment records caused by repeated submissions through idempotency tokens, uniqueness constraints, transactional validation, and request-state verification.
- Implemented caching for frequently requested eligibility, location, and reference data with explicit invalidation, short expiration periods, and monitoring to improve speed without serving stale decisions.
- Developed scheduled reconciliation services for incomplete claims and delayed provider callbacks, recovering transactions after timeout conditions while preserving audit records for service and finance teams.
- Expanded automated coverage with **NUnit**, **Moq**, Selenium, service virtualization, API integration tests, and failure-scenario suites for claims, payments, appointments, and eligibility decisions.

Junior Software Developer
June 2015 - December 2016

Rustici Software — Hybrid / Full time, Nashville, TN

- Maintained learning-platform applications using **Python 3.4-3.5 scripts, C#, ASP.NET MVC 5, MySQL, reporting automation** for course imports, learner-state persistence, launch configuration, completion tracking, reporting extracts, and xAPI-style event handling.
- Implemented administration workflows and integration endpoints for content ingestion, adding validation for malformed archives, incomplete manifests, missing metadata, and nonstandard customer payloads.
- Resolved excessive memory consumption during large imports by streaming compressed packages, limiting parallel extraction, disposing temporary resources, and recording actionable processing failures for support teams.
- Built defensive processing for duplicate and out-of-order learner events, preserving accurate completion, score, and resume state across platforms with inconsistent callback and retry behavior.
- Improved database reliability through targeted indexes, transactional updates, parameterized queries, and repeatable migration scripts for learner-state tables used during high-volume reporting periods.
- Collaborated with senior engineers and support specialists to reproduce unusual integration failures, document payload assumptions, and deliver maintainable fixes that shortened customer troubleshooting cycles.

SELECTED PROJECTS

- **Python API Modernization Platform:** Modernized legacy APIs with **Python 3.8-3.12, Flask 2-3, SQLAlchemy, PostgreSQL, Redis, Celery, Kafka, AWS, Docker**, secure authentication, schema migrations, asynchronous jobs, and observability dashboards that improved maintainability and release confidence.
- **Asynchronous Data Processing Service:** Designed an event-driven workflow platform with durable messaging, idempotent consumers, transactional updates, replay tools, and production runbooks for high-volume business operations.

EDUCATION

Tennessee State University — Bachelor’s Degree in Computer Science Sep 2011 - May 2015

CERTIFICATIONS

- Microsoft Certified: Azure Developer Associate
- Microsoft Certified: Azure Solutions Architect Expert
- AWS Certified Developer - Associate
- Professional Scrum Master I - PSM I